Delfar: Short and Fast Full-Anonymous Ring Signature Using Generalized Δ-KLPT

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Abstract: We consider the fundamental task of computing an isogeny of a prescribed smooth-degree between two supersingular elliptic curves whose endomorphism rings are known. Via the Deuring correspondence, this task can be reduced to the following problem: given two left $\mathcal{O}_0$-ideals I_1 and I_2, find an ideal equivalent to $\overline{I_1}I_2$ with a prescribed norm. Problems of this type are typically addressed using KLPT-type algorithms. In particular, SQIsign employs the SigningKLPT algorithm for this task. More recently, Δ-KLPT was proposed as a new KLPT-type algorithm. While Δ-KLPT allows a smaller target norm than SigningKLPT, and thus leads to more efficient protocols, it imposes the strong restriction that the two input ideals I_1 and I_2 must have the same prime norm, which limits its applicability. In this paper, we construct a new algorithm Δ-KLPT$^+$, a generalization of Δ-KLPT that removes this restriction while preserving its ability to achieve a smaller target norm. For general inputs, while SigningKLPT outputs an equivalent ideal of norm $\tilde{O}(p^{4.5})$, our Δ-KLPT$^+$ outputs one of norm $\tilde{O}(p^4)$. As an application, we present Delfar, a new fully anonymous ring signature scheme based on Δ-KLPT$^+$. Compared with Erebor-full, the previous fully anonymous ring signature scheme based on SigningKLPT, Delfar achieves both shorter signatures and higher computational efficiency.

1. Introduction

Ring Signatures. Ring signatures, introduced by Rivest, Shamir, and Tauman [25], allow a user to sign a message on behalf of a set of public keys while hiding which member of the set produced the signature. Unlike group signatures, ring signatures do not require any trusted setup or group manager. Today, ring signatures are widely used in a variety of applications such as electronic voting systems [19] and blockchain technologies [14], [23].

Numerous ring signature schemes based on classical assumptions have been proposed [2], [6], [16], [25], [29]. However, with the emergence of quantum computers, constructing ring signatures based on post-quantum assumptions has become an important research direction. As a result, various post-quantum ring signature schemes have been proposed [4], [5], [8], [15], [17], [20], [29].

Isogeny-Based Signatures. Isogeny-based cryptography is a promising candidate for post-quantum cryptography due to its compact public keys and signatures. One of the representative isogeny-based signature schemes is SQIsign [11], which was submitted to the NIST PQC standardization process for additional digital signatures [9].

Recent works have introduced more efficient constructions by exploiting higher-dimensional isogeny techniques, such as SQIsignHD [10], SQIsign2D-West [3], SQIsign2D-East [21], SQIPrime [13], SQIsign2D^2 [28], and SQIsign [1] in Round 2 of the NIST competition. For clarity, we hereafter refer to the Round 1 and Round 2 versions of SQIsign as SQIsign1.0 and SQIsign2.0, respectively.

These schemes achieve significant performance improvements by utilizing isogenies of non-smooth degrees. However, this design choice complicates the analysis of their underlying Σ-protocols. In particular, constructing a simulator that generates transcripts containing non-smooth-degree isogenies without access to the secret key is difficult. Consequently, honest-verifier zero-knowledge is typically proven only in the presence of special oracles or hint distributions that return non-smooth-degree isogenies.

Ring Signatures from Isogenies. A natural approach to constructing ring signatures is to apply a general transformation such as AOS framework [2] on Σ-protocols. However, to apply AOS framework, the Σ-protocol must satisfy special HVZK, which means the existence of a PPT simulator that generates transcripts from a public key and a challenge. Therefore, recent Σ-protocols relying on non-smooth-degree isogenies cannot apply this transformation.

For this reason, Σ-protocols based on smooth-degree isogenies remain relevant. Indeed, Erebor [8] is a ring signature scheme obtained by applying the AOS framework to the Σ-protocol of SQIsign1.0 [9], which relies on smooth-degree isogenies.

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Computing Smooth-Degree Isogenies. One of the main challenges in constructing protocols using smooth-degree is the need to generate a smooth-degree isogeny between two elliptic curves E_1, E_2 with known endomorphism rings. Via the Deuring correspondence, this task can be translated into an ideal-theoretic problem:

Given connecting ideals I_i from a special extremal order $\mathcal{O}_0$ to $\mathcal{O}_i \cong \mathrm{End}(E_i)$ for $i = 1, 2$, find an ideal equivalent to $\overline{I_1} I_2$ with a prescribed smooth norm.

Problems of this type are typically addressed using KLPT-type algorithms. For example, in SQIsign1.0, this task is performed by using the SigningKLPT algorithm [11], Algorithm 5, which produces an ideal of norm approximately $p^{4.5}$ in the standard setting, or approximately $p^{3.75}$ with an optimized parameter setting. In the setting of Erebor-full, the fully anonymous Erebor, the output norm of SigningKLPT further increases to approximately $p^{5.25}$, resulting in a significant computational cost.

More recently, Δ-KLPT was proposed in [22] as a new KLPT-type algorithm that achieves a smaller output norm than SigningKLPT. However, Δ-KLPT requires its two input ideals to have the same prime norm, which severely restricts its applicability.

1.1 Contributions

In this paper, we make the following contributions.

- We propose a new algorithm Δ-KLPT$^+$, a generalization of Δ-KLPT. Unlike Δ-KLPT, our algorithm accepts arbitrary left $\mathcal{O}_0$-ideals as input. For general inputs, while SigningKLPT outputs an equivalent ideal of norm $\tilde{O}(p^{4.5})$, our Δ-KLPT$^+$ outputs an equivalent ideal of norm $\tilde{O}(p^4)$. Thus, our algorithm achieves a smaller output norm while allowing the same inputs as SigningKLPT.
- We propose a new isogeny-based ring signature scheme with full anonymity, *Delfar*. By employing Δ-KLPT$^+$, Delfar achieves a smaller signature size and higher computational efficiency than the existing fully anonymous isogeny-based ring signature scheme Erebor-full.

2. Preliminaries

In this section, we summarize some background knowledge used in this paper.

2.1 Notation

Throughout this paper, we use the following notation. We let p be a prime number of cryptographic size, i.e., p is at least about 2^{256} and let λ be a security parameter. For a finite set S, we write $x \xleftarrow{\$} S$ if x is sampled uniformly at random from S. For a positive integer n, we define $[n] := \{1, \ldots, n\}$. For an odd prime q and an integer a, $\left(\frac{a}{q}\right)$ denotes the Legendre symbol.

2.2 Ring Signature

A ring signature scheme allows a signer to anonymously sign a message on behalf of a set of public keys called a *ring*. The verifier is convinced that the signature was produced by one of the ring members but cannot identify which one. A ring signature scheme consists of the following algorithms.

- $\mathsf{KeyGen}(1^\lambda) \to (\mathsf{pk}, \mathsf{sk})$: Generates a public/secret key pair.
- $\mathsf{Sign}(\mathsf{sk}_i, m, R) \to \sigma$: Given a secret key sk_i corresponding to a public key pk_i, a message m, and a ring $R = \{\mathsf{pk}_1, \ldots, \mathsf{pk}_n\}$ containing pk_i, outputs a ring signature σ.
- $\mathsf{Verify}(m, R, \sigma) \to \{0, 1\}$: Outputs 1 if σ is a valid signature on message m with respect to ring R.

In addition, a ring signature must satisfy the following properties.

Correctness. A ring signature scheme is said to be *correct* if for all $(\mathsf{pk}_i, \mathsf{sk}_i)$ generated by KeyGen, all messages m, and all rings R containing pk_i, it holds

$$\Pr[\mathsf{Verify}(m, R, \sigma) = 1] = 1,$$

where $\sigma \leftarrow \mathsf{Sign}(\mathsf{sk}_i, m, R)$.

Unforgeability. A ring signature scheme is said to be *unforgeable with respect to insider corruption* if any probabilistic polynomial-time (PPT) adversary $\mathcal{A}$ has negligible probability of winning the following game played against a challenger.

1. The challenger generates key pairs $(\mathsf{pk}_i, \mathsf{sk}_i) \leftarrow \mathsf{KeyGen}(1^\lambda; \mathsf{r}_i)$ for $i \in [n]$ using random coin r_i and sends $\mathsf{PK} = \{\mathsf{pk}_i\}_{i \in [n]}$ to $\mathcal{A}$.
2. The challenger initializes two empty sets S and C.
3. The adversary $\mathcal{A}$ can access to the following two oracles an arbitrary polynomial number of times:
   - $\mathsf{OSign}(i, m, R)$: If $\mathsf{pk}_i \in R$, it returns $\sigma \leftarrow \mathsf{Sign}(\mathsf{sk}_i, m, R)$ and adds (i, m, R) to S.
   - $\mathsf{OCorrupt}(i)$: It returns r_i and adds pk_i to C.
4. The adversary $\mathcal{A}$ outputs (m^*, R^*, σ^*). If $R^* \subset \mathsf{PK} \backslash C$, $(-, m^*, R^*) \notin S$, and $\mathsf{Verify}(m^*, R^*, \sigma^*) = 1$, $\mathcal{A}$ wins the game.

Anonymity. A ring signature scheme is said to be anonymous if any PPT adversary $\mathcal{A}$ has negligible advantage in the following game played against a challenger.

1. The challenger generates key pairs $(\mathsf{pk}_i, \mathsf{sk}_i) \leftarrow \mathsf{KeyGen}(1^\lambda; \mathsf{r}_i)$ for $i \in [n]$ using random coin r_i and sends $\mathsf{PK} = \{\mathsf{pk}_i\}_{i \in [n]}$ to $\mathcal{A}$.
2. The challenger samples a random bit $b \xleftarrow{\$} \{0, 1\}$.
3. The adversary $\mathcal{A}$ outputs (i_0, i_1, m, R) such that $\mathsf{pk}_{i_0}, \mathsf{pk}_{i_1} \in R$.
4. The challenger computes $\sigma^* \leftarrow \mathsf{Sign}(\mathsf{sk}_{i_b}, m, R)$ and sends σ^* to $\mathcal{A}$.
5. The adversary $\mathcal{A}$ outputs b^*. The advantage of $\mathcal{A}$ is defined as $\mathsf{Adv}(\mathcal{A}) := |\Pr[b^* = b] - 1/2|$.

In addition, a ring signature is said to be *full anonymous*

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if any PPT adversary $\mathcal{A}$ still has negligible advantage in this game where the challenger sends also $\{\mathsf{r}_i\}_{i\in[n]}$ to $\mathcal{A}$ in Step 1.

2.3 Ring Signatures from Σ-Protocols

Ring signatures can be constructed from Σ-protocols using AOS framework [2]. Suppose we have a special sound and special HVZK Σ-protocol $\Pi = (\mathsf{P}_1, \mathsf{P}_2, \mathsf{V})$ for a relation $\mathcal{R}$ with a challenge space $\mathcal{C}$ and a PPT simulator $\mathcal{S}$.

Let $(\mathsf{pk}_i, \mathsf{sk}_i) \in \mathcal{R}$ for $i \in [n]$ and let $R = \{\mathsf{pk}_i\}_{i\in[n]}$ be a ring of public keys. Let $H : \{0,1\}^* \to \mathcal{C}$ be a hash function. Then we obtain a ring signature from Π as follows.

Signing by User U_i

1. $(\mathsf{com}_i, \mathsf{st}_i) \leftarrow \mathsf{P}_1(\mathsf{pk}_i, \mathsf{sk}_i)$.
2. For $j = i+1, \ldots, n, 1, \ldots, i-1$, compute:

$$\mathsf{chl}_j \leftarrow H(m||\mathsf{com}_{j-1}),$$
$$(\mathsf{com}_j, \mathsf{rsp}_j) \leftarrow \mathcal{S}(\mathsf{pk}_j, \mathsf{chl}_j).$$

3. $\mathsf{chl}_i \leftarrow H(m||\mathsf{com}_{i-1})$.
4. $\mathsf{rsp}_i \leftarrow \mathsf{P}_2(\mathsf{pk}_i, \mathsf{sk}_i, \mathsf{st}_i, \mathsf{chl}_i)$.
5. Return $\sigma = (\mathsf{chl}_1, \mathsf{com}_1, \ldots, \mathsf{com}_n, \mathsf{rsp}_1, \ldots, \mathsf{rsp}_n)$.

Verification

1. For $j \in [n]$, let $\mathsf{chl}_{j+1} \leftarrow H(m||\mathsf{com}_j)$.
2. If the following two conditions hold, the verifier accepts.
   (i) $\mathsf{V}(\mathsf{pk}_j, \mathsf{com}_j, \mathsf{chl}_j, \mathsf{rsp}_j) = 1$ for all j,
   (ii) $\mathsf{chl}_{n+1} = \mathsf{chl}_1$.

It is known that, if the underlying Σ-protocol is correct, special sound, and special HVZK, the resulting ring signature is correct, unforgeable, and anonymous. For more detail, see [8], Proposition 2.3 & Proposition 2.4.

2.4 Elliptic Curves and Isogenies

In this paper, we consider supersingular elliptic curves defined over a finite field of characteristic p. Every supersingular elliptic curve has a model defined over $\mathbb{F}_{p^2}$. Therefore, we only consider elliptic curves defined over $\mathbb{F}_{p^2}$.

Basic Facts. An *isogeny* is a non-constant rational map between elliptic curves that is also a group homomorphism. The *degree* of an isogeny φ is its degree as a rational map and is denoted by $\deg\varphi$. When $\deg(\varphi) = \ell$ for an integer ℓ, we say that φ is a *ℓ-isogeny*. An isogeny φ is *separable* if $\#\ker\varphi = \deg\varphi$. A separable isogeny is uniquely determined by its kernel up to post-composition by an isomorphism. For an isogeny $\varphi : E \to E'$, there exists an isogeny $\hat{\varphi} : E' \to E$ called the *dual isogeny*, satisfying

$$\hat{\varphi}\circ\varphi = [\deg\varphi] \quad \text{and} \quad \varphi\circ\hat{\varphi} = [\deg\varphi].$$

Let $\varphi : E \to E_1$ and $\psi : E \to E_2$ be separable isogenies of coprime degrees. If a separable isogeny $\psi' : E_1 \to F$ satisfies $\ker\psi' = \varphi(\ker\psi)$, we say that ψ' is the *push-forward* of ψ by φ and denote it by $\psi' = [\varphi]_*\psi$. Under the same situation, we say that ψ is the *pull-back* of ψ' by φ and denote it by $\psi = [\varphi]^*\psi'$.

For a positive integer ℓ, we denote by $E[\ell]$ the ℓ-torsion subgroup of E. Let E be an elliptic curve defined over $\mathbb{F}_{p^2}$ and let ℓ be an integer such that $E[\ell] \subseteq E(\mathbb{F}_{p^2})$. Then we can compute a deterministic basis of $E[\ell]$ using deterministic algorithms such as [9], Algorithm 3 and [1], Algorithm 2.2.

Computing Isogenies. Let E be an elliptic curve, ℓ a positive integer, and $G \subseteq E$ a subgroup of order ℓ. Then we can compute a separable ℓ-isogeny $\varphi : E \to E'$ with kernel G using Vélu's formulas [26]. More precisely, given E, ℓ, and G, Vélu's formulas allow us to compute the codomain E' in $O(\ell)$ field operations. In addition, given a point $P \in E$, we can compute $\varphi(P)$ in $O(\ell)$ field operations.

When ℓ is smooth, an ℓ-isogeny can be decomposed into a chain of isogenies of small degrees corresponding to the prime factors of ℓ. By computing each isogeny in the chain using Vélu's formulas, we can compute an ℓ-isogeny and evaluate it at a given point in time polynomial in $\log\ell$.

For isogenies of non-smooth degree, recent higher dimensional isogeny techniques provide an alternative approach. In particular, when $\mathrm{End}(E)$ is known, an isogeny of non-smooth degree from E can be computed in polynomial time by embedding the computation into a two-dimensional isogeny.

For an isogeny $\varphi : E \to E'$, we say that information $\mathcal{I}_\varphi$ is an *efficient representation* of φ if, given a point $P \in E$ and $\mathcal{I}_\varphi$, we can compute $\varphi(P)$ in polynomial time. For example, when ℓ is smooth, the tuple (E, ℓ, G) described above provides an efficient representation of an ℓ-isogeny $\varphi : E \to E'$. In this paper, we assume that every isogeny is given by an efficient representation.

2.5 Quaternion Algebras and the Deuring Correspondence

Quaternion Algebras. A *quaternion algebra* over $\mathbb{Q}$ is a division algebra defined by $\mathbb{Q}+\mathbb{Q}\mathbf{i}+\mathbb{Q}\mathbf{j}+\mathbb{Q}\mathbf{k}$ and $\mathbf{i}^2 = a, \mathbf{j}^2 = b, \mathbf{ij} = -\mathbf{ji} = \mathbf{k}$ for $a, b \in \mathbb{Q}^*$. We denote it by $H(a,b)$. We say $H(a,b)$ is *ramified* at a place v of $\mathbb{Q}$ if $H(a,b) \otimes_{\mathbb{Q}} \mathbb{Q}_v$ is not isomorphic to the algebra of the 2×2 matrices over $\mathbb{Q}_v$. There exists a quaternion algebra ramified exactly at p and ∞. Such an algebra is unique up to isomorphism. We denote it by $\mathcal{B}_{p,\infty}$.

Let $\alpha = x+y\mathbf{i}+z\mathbf{j}+t\mathbf{k} \in H(a,b)$ with $x, y, z, t \in \mathbb{Q}$. The *conjugate* of α is $x - y\mathbf{i} - z\mathbf{j} - t\mathbf{k}$ and denoted by $\overline{\alpha}$. The *(reduced) norm* of α is $\alpha\overline{\alpha}$ and denoted by $\mathrm{nrd}(\alpha)$.

A $\mathbb{Z}$-lattice I of rank 4 of $H(a,b)$ is called *ideal*. This means that $I = \mathbb{Z}\alpha_1 + \mathbb{Z}\alpha_2 + \mathbb{Z}\alpha_3 + \mathbb{Z}\alpha_4$ for a basis $\{\alpha_1, \alpha_2, \alpha_3, \alpha_4\}$ of $H(a,b)$. We denote such an ideal by $\mathbb{Z}\langle\alpha_1, \alpha_2, \alpha_3, \alpha_4\rangle$. An ideal $\mathcal{O}$ which is also a subring of $H(a,b)$ is called *order*. An order $\mathcal{O}$ is said to be *maximal* if there is no larger order that contains $\mathcal{O}$.

For an ideal I of $H(a,b)$, we define $\mathcal{O}_L(I) := \{\alpha \in H(a,b) | \alpha\cdot I \subset I\}$, which is known to be an order of $H(a,b)$

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and we call it a *left order* of I. A *right order* $\mathcal{O}_R(I)$ of I is similarly defined. For a maximal order $\mathcal{O}$, an *(integral) left $\mathcal{O}$-ideal* I is an ideal such that $\mathcal{O}_L(I) = \mathcal{O}$ and $I \subset \mathcal{O}$. A *right $\mathcal{O}$-ideal* is similarly defined. When I is a left $\mathcal{O}$-ideal and right $\mathcal{O}'$-ideal, we say that I is a *connecting ideal* from $\mathcal{O}$ to $\mathcal{O}'$.

For an ideal I, we denote its conjugate by $\overline{I} = \{\overline{\alpha} \mid \alpha \in I\}$. We denote by $\mathrm{nrd}(I)$ the *(reduced) norm* of the ideal I, defined as (the unique positive generator of) the $\mathbb{Z}$-module generated by the norms of the elements of I. A left $\mathcal{O}$-ideal I of integer norm can be written as $I = \mathcal{O}\alpha + \mathcal{O}\,\mathrm{nrd}(I)$ for some $\alpha \in I$. The *ideal equivalence* denoted by $I \sim J$ means that there exists $\beta \in H(a,b)^*$ such that $I = J\beta$.

Deuring Correspondence. Deuring [12] showed that the endomorphism ring of a supersingular elliptic curve over $\mathbb{F}_{p^2}$ is isomorphic to a maximal order of $\mathcal{B}_{p,\infty}$ and gave a correspondence called *Deuring correspondence*, where a supersingular elliptic curve E over $\mathbb{F}_{p^2}$ corresponds to a maximal order isomorphic to $\mathrm{End}(E)$.

Suppose $p \equiv 3 \pmod 4$. This is the setting we use in our scheme. Then we can take $\mathcal{B}_{p,\infty} = H(-1,-p)$ and an elliptic curve $E_0/\mathbb{F}_{p^2} : y^2 = x^3 + x$ over $\mathbb{F}_{p^2}$ is supersingular. We define endomorphisms $\iota : (x,y) \mapsto (-x, \sqrt{-1}y)$ and $\pi : (x,y) \mapsto (x^p, y^p)$ of E_0, where $\sqrt{-1}$ is a fixed square root of -1 in $\mathbb{F}_{p^2}$. Then $\mathrm{End}(E_0)$ is isomorphic to $\mathcal{O}_0 := \mathbb{Z}\langle 1, \mathbf{i}, \frac{\mathbf{i}+\mathbf{j}}{2}, \frac{1+\mathbf{k}}{2}\rangle$, which is a *special extremal order* defined in [18]. This isomorphism is given by $\iota \mapsto \mathbf{i}$ and $\pi \mapsto \mathbf{j}$. From now on, we identify $\mathrm{End}(E_0)$ with $\mathcal{O}_0$ by this isomorphism.

The Deuring correspondence also establishes a correspondence between isogenies and ideals. Let $\varphi : E_1 \to E_2$ be a d-isogeny between supersingular elliptic curves over $\mathbb{F}_{p^2}$ and let $\mathcal{O}_1, \mathcal{O}_2$ be maximal orders of $\mathcal{B}_{p,\infty}$ corresponding to E_1, E_2, respectively. Then the isogeny φ can be associated with a connecting ideal I_φ from $\mathcal{O}_1$ to $\mathcal{O}_2$ of norm d defined by

$$I_\varphi := \{\alpha \in \mathcal{O}_1 \mid \alpha(P) = 0,\ \forall P \in \ker\varphi\}.$$

Conversely, for a connecting ideal I from $\mathcal{O}_1$ to $\mathcal{O}_2$ of norm d, we can define the corresponding isogeny $\varphi_I : E_1 \to E_2$ by setting its kernel to

$$E_1[I] := \{P \in E_1 \mid \alpha(P) = 0,\ \forall \alpha \in I\}.$$

Moreover, two isogenies $\varphi, \psi : E_1 \to E_2$ that have the same domain and codomain correspond to equivalent ideals $I_\varphi \sim I_\psi$. For a connecting ideal I from $\mathcal{O}_1$ to $\mathcal{O}_2$ and $\alpha \in I \setminus \{0\}$, we define $\chi_I(\alpha) := I\frac{\overline{\alpha}}{\mathrm{nrd}(I)}$, which is an ideal equivalent to I of norm $\frac{\mathrm{nrd}(\alpha)}{\mathrm{nrd}(I)} \in \mathbb{Z}$.

2.6 Existing Algorithms Related to Deuring Correspondence.

Here, we introduce some existing algorithms using quaternion algebras. As in the above, we let $\mathcal{O}_0$ be the maximal order of $\mathcal{B}_{p,\infty}$ with basis $\{1, \mathbf{i}, \frac{\mathbf{i}+\mathbf{j}}{2}, \frac{1+\mathbf{k}}{2}\}$.

$\mathsf{IdealToIsogeny}(I)$: This is an algorithm to convert an ideal to its corresponding isogeny proposed in [7]. It takes a left $\mathcal{O}_0$-ideal I of odd norm as input and outputs an efficient representation of an isogeny φ_I corresponding to I.

More precisely, this algorithm is designed for the case where p is of the form $p = 2^e \cdot f - 1$, and it returns the evaluation of φ_I on $E_0[2^e]$ as an efficient representation of φ_I. This algorithm internally performs a 2-dimensional isogeny computation.

$\mathsf{StrongApproximation}_M(N, C, D)$: This algorithm was proposed in [18], [24]. It takes as input an integer M, a prime N, and $(C : D) \in \mathbb{P}^1(\mathbb{Z}/N\mathbb{Z})$, and outputs an element $\alpha \in \mathcal{O}_0$ of norm dividing M and of the form $\alpha = u(C\mathbf{j} + D\mathbf{k}) + N\mu_0$, where $u \in \mathbb{Z}$ and $\mu_0 \in \mathcal{O}_0$. For the algorithm to succeed, the integer M must be at least of size $\tilde{O}(pN^3)$. Furthermore, if $\left(\frac{p(C^2+D^2)}{N}\right) = \left(\frac{M}{N}\right)$ holds, this algorithm returns α whose norm is exactly M.

$\mathsf{KLPT}_M(I)$: This algorithm was proposed in [18]. Given a left $\mathcal{O}_0$-ideal I and an integer M, it outputs an equivalent ideal $J \sim I$ of norm M. For this algorithm to succeed, M must be at least of size $\tilde{O}(p^3)$.

$\mathsf{SigningKLPT}_M(I, I')$: This algorithm was proposed as a subroutine of SQIsign-1.0 [11]. It takes as input a left $\mathcal{O}_0$-ideal I of norm N, a left $\mathcal{O}_R(I)$-ideal I' of norm coprime to N, and an integer M. Then it outputs an equivalent ideal $J \sim I'$ of norm M. For this algorithm to succeed, M must be at least of size $\tilde{O}(p^3N^3)$.

When two left $\mathcal{O}_0$-ideals I_1 and I_2 are given, as in the setting considered in this paper, we can apply $\mathsf{SigningKLPT}_M(I_1, \overline{I_1}I_2)$. This yields an equivalent ideal $J \sim \overline{I_1}I_2$ of norm M.

$\Delta\text{-}\mathsf{KLPT}_M(I_1, I_2)$: This algorithm was proposed in [22] as a new KLPT-type algorithm. It takes as input two left $\mathcal{O}_0$-ideals I_1 and I_2 of a same prime norm N and an integer M. Then it outputs an equivalent ideal $I \sim \overline{I_1}I_2$ of norm dividing M. The overview of this algorithm is as follows:

1. Computes $\alpha_0 = C_0\mathbf{j} + D_0\mathbf{k}$ such that $[\alpha_0]_*\varphi_1 = \varphi_2$, where φ_1, φ_2 are the isogenies corresponding to I_1, I_2.
2. Let $\alpha \leftarrow \mathsf{StrongApproximation}_M(N, C_0, D_0)$.
3. Outputs $I = I_1^{-1} \cdot I_2\alpha$, which corresponds to $[\varphi_1]_*\alpha_0$.

Note that this algorithm is based on the following commutative diagram.

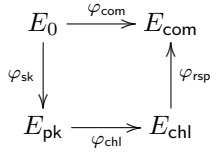
$$\begin{array}{ccc} E_0 & \xrightarrow{\alpha_0} & E_0 \\ {\scriptstyle\varphi_1}\downarrow & & \downarrow{\scriptstyle\varphi_2} \\ E_1 & \xrightarrow[\varphi_I]{} & E_2 \end{array}$$

Since we use $\mathsf{StrongApproximation}$, the integer M must be at least of size $\tilde{O}(pN^3)$. Furthermore, if $\left(\frac{p(C_0^2+D_0^2)}{N}\right) = \left(\frac{M}{N}\right)$ holds, then $\Delta\text{-}\mathsf{KLPT}$ outputs an ideal whose norm is exactly M. Note that the value of $\left(\frac{p(C_0^2+D_0^2)}{N}\right)$ is uniquely determined by the input ideals I_1 and I_2, and is independent of the choice of C_0 and D_0 in Step 1.

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3. Delfar: New Ring Signature Scheme

Here, we introduce our new ring signature scheme Delfar. Our signature is constructed by applying the AOS framework [2] on our new Σ-protocol based on the following diagram.



In this diagram, $(E_{\mathsf{pk}}, \varphi_{\mathsf{sk}})$ is a key pair and E_{com} is a commitment. Against a challenge isogeny $\varphi_{\mathsf{chl}} : E_{\mathsf{pk}} \to E_{\mathsf{chl}}$, the prover computes an isogeny $\varphi_{\mathsf{rsp}} : E_{\mathsf{pk}} \to E_{\mathsf{com}}$ of a fixed degree $M := 2^{e_{\mathsf{rsp}}}$. From now on, let I_1, I_2 be the left $\mathcal{O}_0$-ideals corresponding to $\varphi_{\mathsf{chl}} \circ \varphi_{\mathsf{sk}}$ and φ_{com}, respectively.

Our Σ-protocol is essentially the same as that of Erebor, which is itself a slight modification of the Σ-protocol of SQIsign1.0. The main idea is to use $\Delta\text{-}\mathsf{KLPT}_M$ instead of $\mathsf{SigningKLPT}_M$ to compute the ideal corresponding to φ_{rsp}. However, directly using $\Delta\text{-}\mathsf{KLPT}_M$ introduces two major challenges.

1. For $\Delta\text{-}\mathsf{KLPT}_M$ to output an ideal of norm exactly M, the input ideals I_1 and I_2 must satisfy the following conditions.

$$\mathrm{nrd}(I_1) = \mathrm{nrd}(I_2)(=: N), \tag{1}$$

$$\text{The common norm } N \text{ is prime}, \tag{2}$$

$$\left(\frac{p(C_0^2 + D_0^2)}{N}\right) = \left(\frac{M}{N}\right), \tag{3}$$

   where C_0 and D_0 are the coefficients obtained in Step 1 of $\Delta\text{-}\mathsf{KLPT}_M(I_1, I_2)$. However, it is difficult to design the protocol such that I_1 and I_2 satisfy these conditions. In particular, $\mathrm{nrd}(I_1) = \deg(\varphi_{\mathsf{chl}} \circ \varphi_{\mathsf{sk}})$ can clearly never be prime.
2. Let φ_{rsp} be the isogeny computed via $\Delta\text{-}\mathsf{KLPT}$. Then the isogeny obtained by pulling back φ_{rsp} along $\varphi_{\mathsf{chl}} \circ \varphi_{\mathsf{sk}}$ is always an endomorphism of E_0, which prevents the resulting ring signature scheme from achieving full anonymity.

To address these two issues, we introduce a new algorithm, $\Delta\text{-}\mathsf{KLPT}^+$, which generalizes $\Delta\text{-}\mathsf{KLPT}$ to arbitrary pairs of left $\mathcal{O}_0$-ideals. Specifically, $\Delta\text{-}\mathsf{KLPT}^+$ takes as input two arbitrary left $\mathcal{O}_0$-ideals I_1 and I_2, together with an integer M, and outputs an ideal equivalent to $\overline{I_1} I_2$ whose norm is M.

3.1 New Algorithm: $\Delta\text{-}\mathsf{KLPT}^+$

At a high level, $\Delta\text{-}\mathsf{KLPT}^+$ consists of the following four phases:

1. **Norm Alignment.** Using KLPT, we transform the two input ideals I_1 and I_2 into equivalent ideals of the same norm, thereby satisfying condition 1.
2. **Randomization.** We randomize the pair of ideals while preserving the equality of their norms. This phase provides the randomness necessary for achieving full anonymity.
3. **Norm Reduction.** We reduce the common norm of the ideal pair. This phase is essential for improving the efficiency of our protocol. After reducing the norm, we check whether the resulting ideal pair satisfies the remaining input requirements of $\Delta\text{-}\mathsf{KLPT}_M$, namely, conditions 2 and 3. If not, we return to the randomization phase.
4. **Δ-KLPT.** Finally, we apply $\Delta\text{-}\mathsf{KLPT}_M$ to the resulting pair to compute an ideal equivalent to $\overline{I_1} I_2$ of norm M.

We now describe each phase in detail.

Phase 1: Norm Alignment. This phase can be realized straightforwardly using KLPT. More precisely, we first fix an integer T of size approximately p^3. We then apply KLPT_T independently to I_1 and I_2 to obtain equivalent ideals L_1 and L_2 satisfying $\mathrm{nrd}(L_1) = T$ and $\mathrm{nrd}(L_2) = T$.

Phase 2: Randomization. We sample $r \xleftarrow{\$} [0, p]$ and set $\gamma = 1 + r\mathbf{i}$ if $r < p$, and $\gamma = \mathbf{i}$ otherwise. We then set $J_1 = L_1\gamma$ and $J_2 = L_2\overline{\gamma}$. Then we have $\mathrm{nrd}(J_1) = \mathrm{nrd}(J_2) = T\,\mathrm{nrd}(\gamma)$. The pairs of same-norm ideals (J_1, J_2) equivalent to (L_1, L_2) can be partitioned into $(p+1)/2$ classes. The above randomization covers these classes uniformly; hence, the resulting pair (J_1, J_2) belongs to a uniformly random class among the $(p+1)/2$ classes.

Phase 3: Norm Reduction. This phase is the main technical challenge. To realize this phase, we construct a new subroutine $\mathsf{IdealNormReduce}$. This subroutine, described in Algorithm 1, takes as input two left $\mathcal{O}_0$-ideals I_1 and I_2 of the same norm N and outputs equivalent ideals I_1' and I_2' of the same norm N', where N' is smaller than $c \cdot p^{3/4} N^{1/4}$ for a constant $c < 1$. Our construction is based on the following lemmas. The proofs are given in Appendix.

Lemma 1. *Let I_1 and I_2 be left $\mathcal{O}_0$-ideals of the same norm N. Then, $\mathcal{O}_0 \cap I_2^{-1} I_1$ is a lattice in $\mathcal{O}_0$ and $[\mathcal{O}_0 : \mathcal{O}_0 \cap I_2^{-1} I_1] \leq N$.*

Lemma 2. *Let I_1 and I_2 be left $\mathcal{O}_0$-ideals of the same norm N, and $x \in \mathcal{O}_0 \cap I_2^{-1} I_1$ be a nonzero element. Suppose that I_1 and I_2 are primitive and p does not divide N. Then, we have $[\mathcal{O}_0 : I_1 \cap x^{-1} I_2 x] \leq N^2 \cdot \mathrm{nrd}(x)$.*

In Step 1, we choose a shortest nonzero vector $x \in \mathcal{O}_0 \cap I_2^{-1} I_1$. By Lemma 1 and Minkowski's theorem, we have

$$\mathrm{nrd}(x) \leq \frac{2\sqrt{2pN}}{\pi}.$$

Next, in Step 2, we choose a shortest nonzero vector $\beta_1 \in I_1 \cap x^{-1} I_2 x$. By Lemma 2 and Minkowski's theorem, we have

$$\mathrm{nrd}(\beta_1) \leq \frac{2\sqrt{2pN^2\,\mathrm{nrd}(x)}}{\pi} \leq c \cdot p^{3/4} N^{5/4},$$

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Algorithm 1: $\mathsf{IdealNormReduce}(I_1, I_2)$

Input: Left $\mathcal{O}_0$-ideals I_1, I_2 of the same norm N.
Output: Equivalent ideals I_1', I_2' of the same norm $N' \leq c \cdot p^{3/4} N^{1/4}$.

1 Choose a shortest nonzero vector $x \in \mathcal{O}_0 \cap I_2^{-1} I_1$;
2 Choose a shortest nonzero vector $\beta_1 \in I_1 \cap x^{-1} I_2 x$;
3 $\beta_2 \leftarrow x\beta_1 x^{-1}$;
4 $I_1' \leftarrow \chi_{I_1}(\beta_1)$ and $I_2' \leftarrow \chi_{I_2}(\beta_2)$;
5 **return** (I_1', I_2');

where $c = \left(\frac{2\sqrt{2}}{\pi}\right)^{3/2} \approx 0.854$.

Finally, in Step 3, we let $\beta_2 := x\beta_1 x^{-1} \in I_2$, and return $I_1' := \chi_{I_1}(\beta_1)$ and $I_2' := \chi_{I_2}(\beta_2)$. Since $\mathrm{nrd}(\beta_2) = \mathrm{nrd}(x\beta_1 x^{-1}) = \mathrm{nrd}(\beta_1)$, the output ideals have the same norm $N' = \frac{\mathrm{nrd}(\beta_1)}{N} \leq c \cdot p^{3/4} N^{1/4}$.

We now show that, by applying $\mathsf{IdealNormReduce}$ repeatedly, the common norm of the ideal pair can be reduced below p. Let $(I_1^{(0)}, I_2^{(0)})$ be an ideal pair of common norm N_0, and, for each $i \geq 0$, let

$$(I_1^{(i+1)}, I_2^{(i+1)}) \leftarrow \mathsf{IdealNormReduce}(I_1^{(i)}, I_2^{(i)}).$$

Writing N_i for the common norm of $I_1^{(i)}$ and $I_2^{(i)}$, the above analysis gives

$$N_{i+1} \leq c \cdot p^{3/4} N_i^{1/4}.$$

By iterating this inequality, we obtain

$$N_k \leq c^{\frac{4}{3}(1-4^{-k})} p^{1-4^{-k}} N_0^{4^{-k}} < \left(c^{\frac{4}{3}(1-4^{-k})} N_0^{4^{-k}}\right) p.$$

Thus, by taking $k \geq \log_4 \left(\frac{3\log(N_0/c)}{4\log(1/c)}\right)$, we have $N_k < p$.

Therefore, after $O(\log\log N_0)$ applications of $\mathsf{IdealNormReduce}$, the common norm can be reduced below p.

Phase 4: Δ-KLPT. Let (J_1, J_2) denote the ideal pair obtained after the norm-reduction phase, and let N be their common norm. We apply $\Delta\text{-}\mathsf{KLPT}_M(J_1, J_2)$ to obtain an ideal J of norm M. Since J_1 and J_2 are equivalent to the original input ideals I_1 and I_2, respectively, the resulting ideal J is equivalent to $\overline{I_1} I_2$.

Recall that $\Delta\text{-}\mathsf{KLPT}_M$ requires $M = \tilde{O}(pN^3)$. Since the norm-reduction phase ensures that $N < p$, it suffices to take $M = \tilde{O}(p^4)$. Therefore, $\Delta\text{-}\mathsf{KLPT}^+{}_M$ outputs an ideal J equivalent to $\overline{I_1} I_2$ with norm $M = \tilde{O}(p^4)$.

3.2 Underlying Σ-Protocol

Now we describe our Σ-protocol underling Delfar in detail.

Parameter setting. For the security parameter λ, the public parameter is set as follows:

- A large prime $p = 2^e f - 1$ with $e \approx 2\lambda$ and $f \in \mathbb{N}$ as small as possible.
- An elliptic curve $E_0 : y^2 = x^3 + x$ over $\mathbb{F}_{p^2}$.
- A maximal order $\mathcal{O}_0 = \mathbb{Z}\langle 1, \mathbf{i}, \frac{\mathbf{i}+\mathbf{j}}{2}, \frac{1+\mathbf{k}}{2}\rangle$ of $B_{p,\infty}$. This order is isomorphic to $\mathrm{End}(E_0)$ and we identify $\mathrm{End}(E_0)$ with $\mathcal{O}_0$ via the isomorphism described in § 2.5.
- An integer $M_{\mathsf{rsp}} = 2^{e_{\mathsf{rsp}}} = 2^{4e+\delta} \approx p^4$, where δ is an additional parameter that provides sufficient margin for $\Delta\text{-}\mathsf{KLPT}^+{}_{M_{\mathsf{rsp}}}$ to succeed. It suffices to take $\delta = O(\log\log\lambda)$; in particular, we can choose δ so that $\delta \leq e - \lambda \approx \lambda$.
- A prime $D \approx 2^{2\lambda}$, which is used as the degree of φ_{sk} and φ_{com}.

Algorithm 2: $\Delta\text{-}\mathsf{KLPT}^+{}_M(I_1, I_2)$

Input: Left $\mathcal{O}_0$-ideals I_1, I_2 and an integer M.
Output: An ideal J equivalent to $\overline{I_1} I_2$ of norm M.

// Phase 1: Norm Alignment
1 Let T be a fixed integer of size approximately p^3;
2 $L_1 \leftarrow \mathsf{KLPT}_T(I_1)$;
3 $L_2 \leftarrow \mathsf{KLPT}_T(I_2)$;
4 **repeat**
  // Phase 2: Randomization
5 Sample $r \xleftarrow{\$} [0, p]$;
6 **if** $r < p$ **then**
7 $\gamma \leftarrow 1 + r\mathbf{i}$;
8 **else**
9 $\gamma \leftarrow \mathbf{i}$;
10 $J_1 \leftarrow L_1\gamma$;
11 $J_2 \leftarrow L_2\overline{\gamma}$;
  // Phase 3: Norm Reduction
12 **while** $\mathrm{nrd}(J_1) \geq p$ **do**
13 $(J_1, J_2) \leftarrow \mathsf{IdealNormReduce}(J_1, J_2)$;
14 **until** (J_1, J_2) *satisfies Conditions 2 and 3*;
// Phase 4: Δ-KLPT
15 $J \leftarrow \Delta\text{-}\mathsf{KLPT}_M(J_1, J_2)$;
16 **return** J;

Key generation. In the key generation phase, we compute a random D-isogeny $\varphi_{\mathsf{sk}} : E_0 \to E_{\mathsf{pk}}$ and its corresponding ideal I_{sk}. More precisely, we choose a random left $\mathcal{O}_0$-ideal I_{sk} of norm D and compute φ_{sk} using $\mathsf{IdealToIsogeny}$. We set $\mathsf{pk} = E_{\mathsf{pk}}$ and $\mathsf{sk} = (\varphi_{\mathsf{sk}}, I_{\mathsf{sk}})$.

Commitment. As in the key generation phase, we compute a random D-isogeny $\varphi_{\mathsf{com}} : E_0 \to E_{\mathsf{com}}$ and its corresponding ideal I_{com} using $\mathsf{IdealToIsogeny}$. We set $\mathsf{com} = E_{\mathsf{com}}$ and $\mathsf{st} = (\varphi_{\mathsf{com}}, I_{\mathsf{com}})$.

Challenge. We use $\mathcal{C} := \mathbb{Z}/2^\lambda\mathbb{Z}$ as the challenge space. Given a challenge $\mathsf{chl} \xleftarrow{\$} \mathcal{C}$, we define a 2^λ-isogeny $\varphi_{\mathsf{chl}} : E_{\mathsf{pk}} \to E_{\mathsf{chl}}$ by $\ker(\varphi_{\mathsf{chl}}) = \langle P_{\mathsf{pk}} + [\mathsf{chl}]Q_{\mathsf{pk}}\rangle$, where $(P_{\mathsf{pk}}, Q_{\mathsf{pk}})$ is a deterministic basis of $E_{\mathsf{pk}}[2^\lambda]$.

Response. We first compute the left $\mathcal{O}_0$-ideal I_{chl} corresponding to φ_{chl}, and let $I_1 = I_{\mathsf{sk}} I_{\mathsf{chl}}$ and $I_2 = I_{\mathsf{com}}$. We then compute $J_{\mathsf{rsp}} \leftarrow \Delta\text{-}\mathsf{KLPT}^+{}_{M_{\mathsf{rsp}}}(I_1, I_2)$. Finally, we compute an M_{rsp}-isogeny $\varphi_{\mathsf{rsp}} : E_{\mathsf{chl}} \to E_{\mathsf{com}}$ corresponding to J_{rsp}.

We now describe in more detail how to compute an efficient representation of the response isogeny φ_{rsp} corresponding to J_{rsp}. We first factor J_{rsp} as

$$J_{\mathsf{rsp}} = J_\delta J_1 J_2 J_3 J_4,$$

where $\mathrm{nrd}(J_\delta) = 2^\delta$ and $\mathrm{nrd}(J_i) = 2^e$ for $i \in \{1, 2, 3, 4\}$. Let $\varphi_\delta, \varphi_1, \varphi_2, \varphi_3, \varphi_4$ denote the isogenies corresponding to

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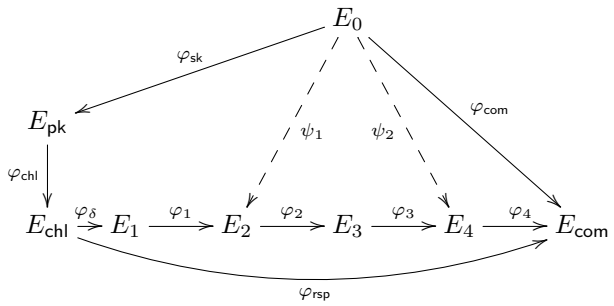


Fig. 1 Overview of the computation of the response isogeny.

$J_\delta, J_1, J_2, J_3, J_4$, respectively. For clarity, we illustrate the overall procedure in Figure 1. Our goal is to compute generators of $\ker(\varphi_\delta)$ and $\ker(\varphi_i)$ for $i \in \{1,2,3,4\}$.

We first compute a generator K_δ of $\ker(\varphi_\delta)$. This can be done relatively easily. More precisely, we compute a generator of $E_0[I_1 J_\delta] \cap E_0[2^{\lambda+\delta}]$ and evaluate it under $\varphi_{\mathsf{chl}} \circ \varphi_{\mathsf{sk}}$. The resulting point K_δ generates $\ker(\varphi_\delta)$. Since $\delta \le e - \lambda$, we have $E_0[2^{\lambda+\delta}] \subseteq E_0[2^e] \subseteq E_0(\mathbb{F}_{p^2})$. Therefore, the computation of K_δ can be performed entirely over $\mathbb{F}_{p^2}$.

We next compute generators of $\ker(\varphi_i)$ for $i \in \{1,2,3,4\}$. To this end, we choose a left $\mathcal{O}_0$-ideal L_1 of odd norm that is equivalent to $I_1 J_\delta J_1$, and a left $\mathcal{O}_0$-ideal L_2 of odd norm that is equivalent to $I_2 \overline{J_4}$. Using $\mathsf{IdealToIsogeny}$, we compute the isogenies ψ_1 and ψ_2 corresponding to L_1 and L_2, respectively.

Using ψ_1 and ψ_2, and proceeding similarly to the computation of K_δ, we obtain generators K_1', K_2, K_3', K_4 of $\ker(\widehat{\varphi_1}), \ker(\varphi_2), \ker(\widehat{\varphi_3}), \ker(\varphi_4)$, respectively. For example, by computing a generator of $E_0[L_1 \overline{J_1}] \cap E_0[2^\lambda]$ and evaluating it under ψ_1, we obtain K_1'.

It remains to recover generators of $\ker(\varphi_1)$ and $\ker(\varphi_3)$. From K_1' and K_3', we first compute the dual isogenies $\widehat{\varphi_1}$ and $\widehat{\varphi_3}$, respectively. For each $i \in \{1,3\}$, we then choose $P_i \in E_{i+1}[2^e]$ that is linearly independent of K_i'. We set $K_i := \widehat{\varphi_i}(P_i)$. Then K_i generates $\ker(\varphi_i)$. Hence, we obtain generators K_1, K_2, K_3, K_4 of $\ker(\varphi_1), \ker(\varphi_2), \ker(\varphi_3), \ker(\varphi_4)$, respectively.

Finally, we return $\mathsf{rsp} = (K_\delta, K_1, K_2, K_3, K_4)$ as an efficient representation of φ_{rsp}.

Verification. Given a commitment E_{com}, a challenge chl, and a response $\mathsf{rsp} = (K_\delta, K_1, K_2, K_3, K_4)$, we first compute a 2^λ-isogeny φ_{chl} determined by chl and its codomain E_{chl}. Then we reconstruct the response isogeny φ_{rsp} from rsp using Vélu's formulas. We accept if the resulting φ_{rsp} is a valid M_{rsp}-isogeny from E_{chl} to E_{com}, and reject otherwise.

3.3 Resulting Ring Signature Scheme

We now construct a new ring signature scheme, Delfar, by applying the AOS framework [2] to the Σ-protocol proposed in § 3.2.

To apply the AOS framework, we need to construct a PPT simulator $\mathcal{S}$ that, given a public key pk and a challenge chl, outputs a commitment-response pair $(\mathsf{com}, \mathsf{rsp})$ forming a valid transcript $(\mathsf{com}, \mathsf{chl}, \mathsf{rsp})$. We consider the following simulator.

- $\mathcal{S}(\mathsf{pk} := E_{\mathsf{pk}}, \mathsf{chl}) \to (\mathsf{com}, \mathsf{rsp})$:
  1. Let $(P_{\mathsf{pk}}, Q_{\mathsf{pk}})$ be a deterministic basis of $E_{\mathsf{pk}}[2^\lambda]$.
  2. Compute the 2^λ-isogeny $\varphi_{\mathsf{chl}} : E_{\mathsf{pk}} \to E_{\mathsf{chl}}$ such that $\ker(\varphi_{\mathsf{chl}}) = \langle P_{\mathsf{pk}} + [\mathsf{chl}]Q_{\mathsf{pk}} \rangle$.
  3. Sample an M_{rsp}-isogeny $\varphi_{\mathsf{rsp}} : E_{\mathsf{chl}} \to E_{\mathsf{com}}$ uniformly at random among all M_{rsp}-isogenies with domain E_{chl}.
  4. Let $\mathsf{com} = E_{\mathsf{com}}$ and $\mathsf{rsp} = (K_\delta, K_1, \ldots, K_4)$ be the representation of φ_{rsp} described in § 3.2.

By applying the AOS framework on our Σ-protocol along with the above simulator $\mathcal{S}$, we obtain a new ring signature scheme Delfar.

4. Efficiency

In this section, we evaluate the efficiency of Delfar and compare it with Erebor-full.

4.1 Signature Sizes

The signature σ of Delfar for a ring of n parties consists of $\sigma = (\mathsf{chl}_1, \{\mathsf{com}_j\}_{j\in[n]}, \{\mathsf{rsp}_j\}_{j\in[n]})$, where each com_j is a curve $E_{\mathsf{com},j}$ and each rsp_j represents a $2^{e_{\mathsf{rsp}}}$-isogeny $\varphi_{\mathsf{rsp},j} : E_{\mathsf{chl},j} \to E_{\mathsf{com},j}$ for $j \in [n]$. Since $E_{\mathsf{com},j}$ is the codomain of $\varphi_{\mathsf{rsp},j}$, it can be recovered from rsp_j. Hence, the curves $\{E_{\mathsf{com},j}\}_{j\in[n]}$ do not need to be explicitly included in the signature, and σ can be represented as $\sigma = (\mathsf{chl}_1, \mathsf{rsp}_1, \ldots, \mathsf{rsp}_n)$.

As described in [9], Section 8.5, each response rsp_j, representing a $2^{e_{\mathsf{rsp}}}$-isogeny, can be compressed into $e_{\mathsf{rsp}} + 4(\lceil e_{\mathsf{rsp}}/e \rceil - 1)$ bits. Since chl_1 is represented by λ bits, the total signature size is $\lambda + n\,(e_{\mathsf{rsp}} + 4(\lceil e_{\mathsf{rsp}}/e \rceil - 1)) = \lambda + n(4e + \delta + 16) \approx (8n+1)\lambda$ bits, where the approximation assumes that δ is sufficiently small compared with λ. For comparison, the signature size of Erebor-full is approximately $(10.5n+1)\lambda$ bits. Therefore, Delfar reduces the signature size from approximately $(10.5n+1)\lambda$ bits to $(8n+1)\lambda$ bits.

4.2 Computational Cost

Signing Cost. By the construction of the AOS framework, signing requires one commitment generation, one response generation, and $n-1$ executions of the simulator $\mathcal{S}$ for a ring of n parties. Therefore, when n is sufficiently large, the signing cost is dominated by the executions of $\mathcal{S}$.

In both Erebor-full and Delfar, each execution of $\mathcal{S}$ requires the computation of a 2^λ-isogeny and a $2^{e_{\mathsf{rsp}}}$-isogeny. The main difference lies in the size of e_{rsp}: it is approximately 10.5λ in Erebor-full, whereas it is approximately 8λ in Delfar. Consequently, the total length of the 2-isogeny chains computed in each execution of $\mathcal{S}$ is approximately 11.5λ for Erebor-full and 9λ for Delfar. Thus, Delfar reduces the dominant signing cost compared with Erebor-full.

Verification Cost. Verification in the AOS framework essentially consists of performing the verification procedure of the

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underlying Σ-protocol for each of the n ring members. For both Erebor-full and Delfar, each verification requires the computation of a 2^{λ}-isogeny and a $2^{e_{\mathrm{rsp}}}$-isogeny. Hence, the total length of the 2-isogeny chains per verification is again approximately 11.5λ for Erebor-full and 9λ for Delfar. Since this procedure is performed n times, Delfar also achieves a lower verification cost than Erebor-full.

Our comparison above is based on the lengths of the isogeny chains involved in the signing and verification procedures. A detailed implementation and experimental evaluation of Delfar is left for future work.

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Appendix

For a prime q, we denote the localization of a lattice in $\mathcal{O}_0$ at q by $L_q \coloneqq L \otimes_{\mathbb{Z}} \mathbb{Z}_q$.

A.1 Proof of Lemma 1

Proof. It is clear that $\mathcal{O}_0 \cap I_2^{-1} I_1$ is a lattice in $\mathcal{O}_0$.

From the local-global principle [27], Lemma 9.6.7, it suffices to show that

$$[(\mathcal{O}_0)_q : (\mathcal{O}_0 \cap I_2^{-1} I_1)_q] \leq q^{v_q(N)},$$

where $v_q(N)$ is the q-adic valuation of N. If q does not divide N, then the inequality is clear since $(I_1)_q = (I_2)_q = (\mathcal{O}_0)_q$. If $q = p$, then $(I_1)_p = (I_2)_p = \mathfrak{P}^{v_p(N)}$, where $\mathfrak{P}$ is the unique maximal ideal of $(\mathcal{O}_0)_p$. Therefore, we have $[(\mathcal{O}_0)_p : (\mathcal{O}_0 \cap I_2^{-1} I_1)_p] = 1$, and the inequality holds.

Let $q \neq p$ be a prime dividing N, and $e \coloneqq v_q(N)$. Then, we have $(\mathcal{O}_0)_q \cong \mathrm{M}_2(\mathbb{Z}_q)$. Under this isomorphism, we can write

$$(I_1)_q = \mathrm{M}_2(\mathbb{Z}_q) m_1, \quad (I_2)_q = \mathrm{M}_2(\mathbb{Z}_q) m_2,$$

where $m_1, m_2 \in \mathrm{M}_2(\mathbb{Z}_q)$ are matrices of determinant q^e. By the theory of elementary divisors, there exist $u_1, u_2, v_1, v_2 \in \mathrm{GL}_2(\mathbb{Z}_q)$ and $0 \leq f_1, f_2 \leq e$ such that

$$m_1 = u_1 \begin{pmatrix} q^{f_1} & 0 \\ 0 & q^{e-f_1} \end{pmatrix} v_1, \quad m_2 = u_2 \begin{pmatrix} q^{f_2} & 0 \\ 0 & q^{e-f_2} \end{pmatrix} v_2.$$

Therefore, we have

$$\begin{aligned} &(\mathcal{O}_0 \cap I_2^{-1} I_1)_q \\ &\cong \{\mu \in \mathrm{M}_2(\mathbb{Z}_q) \mid m_2 \mu m_1^{-1} \in \mathrm{M}_2(\mathbb{Z}_q)\} \\ &= \left\{\mu \in \mathrm{M}_2(\mathbb{Z}_q) \;\middle|\; u_2 \mathbf{A}_2 v_2 \mu v_1^{-1} \mathbf{A}_1^{-1} u_1^{-1} \in \mathrm{M}_2(\mathbb{Z}_q)\right\}, \end{aligned}$$

where $\mathbf{A}_i = \begin{pmatrix} q^{f_i} & 0 \\ 0 & q^{e-f_i} \end{pmatrix}$ for $i = 1, 2$. This implies that $\mu \in (\mathcal{O}_0 \cap I_2^{-1} I_1)_q$ if and only if

$$v_2 \mu v_1^{-1} = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

where $a, b, c, d \in \mathbb{Z}_q$ satisfy $v_q(a) \geq f_1 - f_2$, $v_q(b) \geq e - f_1 - f_2$, $v_q(c) \geq f_1 + f_2 - e$, and $v_q(d) \geq f_2 - f_1$. Therefore, we have

$$[(\mathcal{O}_0)_q : (\mathcal{O}_0 \cap I_2^{-1} I_1)_q] = q^{|f_1 - f_2| + |e - f_1 - f_2|}.$$

If $f_1 + f_2 \leq e$, then $|f_1 - f_2| + |e - f_1 - f_2| = e - 2\min(f_1, f_2) \leq e$. If $f_1 + f_2 > e$, then $|f_1 - f_2| + |e - f_1 - f_2| = 2\max(f_1, f_2) - e \leq e$. In either case, we have $|f_1 - f_2| + |e - f_1 - f_2| \leq e$. □

A.2 Proof of Lemma 2

Proof. Since $[\mathcal{O}_0 : I_1] = N^2$, the claim is equivalent to $[I_1 : I_1 \cap x^{-1} I_2 x] \leq \mathrm{nrd}(x)$. By the local-global principle, it suffices to show that

$$[(I_1)_q : (I_1 \cap x^{-1} I_2 x)_q] \leq q^{v_q(\mathrm{nrd}(x))}.$$

In the case where $q = p$, we have $(I_1)_p$ and $(I_2)_p$ are equal to the unique maximal order $(\mathcal{O}_0)_p$ of $\mathcal{B}_{p,\infty} \otimes_{\mathbb{Q}} \mathbb{Q}_p$, Therefore, we have $[(I_1)_p : (I_1 \cap x^{-1} I_2 x)_p] = 1$.

Let q be a prime not dividing $\mathrm{nrd}(x)$. Then, x is a unit in $(\mathcal{O}_0)_q$, and we have $(x^{-1} I_2 x)_q = (I_1)_q$ since $I_2 x \subset I_1$. Therefore, $[(I_1)_q : (I_1 \cap x^{-1} I_2 x)_q] = 1$.

Let $q \neq p$ be a prime dividing $\mathrm{nrd}(x)$, and $e \coloneqq v_q(\mathrm{nrd}(x))$. As in the proof of Lemma 1, we identify $(\mathcal{O}_0)_q$ with $\mathrm{M}_2(\mathbb{Z}_q)$ and write

$$(I_1)_q = \mathrm{M}_2(\mathbb{Z}_q) m_1, \quad (I_2)_q = \mathrm{M}_2(\mathbb{Z}_q) m_2.$$

We set $z \coloneqq m_2 x m_1^{-1}$. Since $I_2 x \subset I_1$, we have $z \in \mathrm{M}_2(\mathbb{Z}_q)$. In addition, we have $v_q(\det z) = e$.

By the theory of elementary divisors, there exist $u_1, u_2, v_1, v_2 \in \mathrm{GL}_2(\mathbb{Z}_q)$ and integers $0 \leq a, b \leq e$ such that

$$x = u_1 \begin{pmatrix} q^a & 0 \\ 0 & q^{e-a} \end{pmatrix} v_1, \quad z = u_2 \begin{pmatrix} q^b & 0 \\ 0 & q^{e-b} \end{pmatrix} v_2.$$

Then, we have

$$\begin{aligned} &(I_1 \cap x^{-1} I_2 x)_q \\ &\cong \{\mu m_1 \mid \mu \in \mathrm{M}_2(\mathbb{Z}_q),\ x \mu m_1 x^{-1} \in \mathrm{M}_2(\mathbb{Z}_q) m_2\} \\ &= \{\mu m_1 \mid x \mu z^{-1} \in \mathrm{M}_2(\mathbb{Z}_q)\} \\ &= \left\{\mu m_1 \;\middle|\; \mathbf{A} v_1 \mu v_2^{-1} \mathbf{B}^{-1} \in \mathrm{M}_2(\mathbb{Z}_q)\right\}, \end{aligned}$$

where $\mathbf{A} = \begin{pmatrix} q^a & 0 \\ 0 & q^{e-a} \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} q^b & 0 \\ 0 & q^{e-b} \end{pmatrix}$.

By the same argument as in the proof of Lemma 1, we have $[(I_1)_q : (I_1 \cap x^{-1} I_2 x)_q] \leq q^e$. □

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